

CS-1180 Lab Problem 4: Fibonacci

PURPOSE: Utilization of loops and conditional statements.

DIRECTIONS: The Fibonacci series is a sequence of numbers that appears often in various fields. The first two numbers in the series are 1, and then the next number is the sum of the two previous numbers. For example:

1, 1

$1 + 1 = 2$, so the third number in the series is 2

1, 1, 2

$1 + 2 = 3$, so the next number in the series is 3

1, 1, 2, 3

The first ten Fibonacci numbers are:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55

Write a program that prompts the user for a number n and prints out the first n Fibonacci numbers. If the user enters a negative or zero value for n , display the message:

```
Error: n cannot be negative or zero
```

It is acceptable if your program crashes or misbehaves when the user inputs a non-integer, though a little additional work should allow your program to survive that as well. A challenge!

For full credit, ensure that your program is well commented to explain your code.

EXAMPLE:

```
What is the value of n?
```

```
> 1
```

```
1
```

```
> 2
```

```
1, 1
```

```
> 6
```

```
1, 1, 2, 3, 5, 8
```

```
What is the value of n?
```

```
> -4
```

```
Error: n cannot be negative or zero
```

RUBRIC:

(3 pts) Base Functionality

[1] Reads input correctly

[1] Output is correct

[1] Handles negative input correctly

(1 pts) Style

[1] The code is meaningfully commented

IMPORTANT NOTES:

- Submissions that do not compile will receive a zero
- Submissions with improperly cited AI will receive a zero and an academic integrity violation
- Submissions that are partially or fully copied from another submission will receive a zero and an academic integrity violation